8.1 Stationary Waves

Question Paper

Course	CIE A Level Physics
Section	8. Superposition
Topic	8.1 Stationary Waves
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

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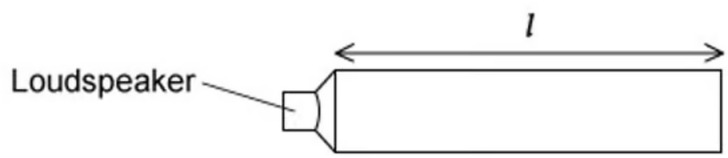
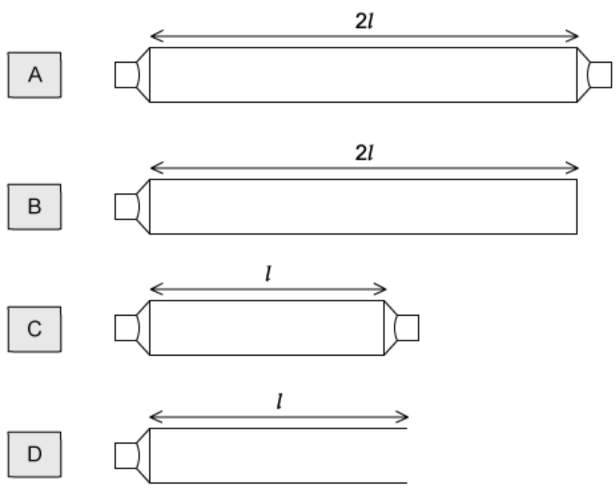
Question 1

A demonstration of a standing wave is set up using a loudspeaker. The loudspeaker is emitting sound with a frequency *f* and is placed at one end of the pipe with length *l*. The pipe is closed at the other end. This is shown in the diagram below.



Different pipes were set up with either one or two loudspeakers of frequency *f*. The pairs of loudspeakers were in phase with each other.

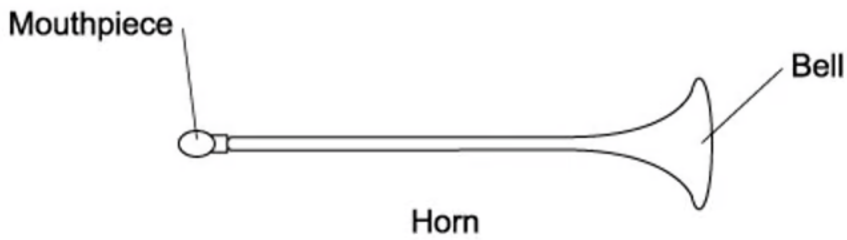
Which pipe would contain a standing wave?



[1 mark]

Question 2

When a wind instrument is used to create a note, a stationary wave is produced in the column. An example of this is the horn.



For the range of notes produced by the horn, a node is formed at the mouthpiece and an antinode is formed at the bell. The frequency of the lowest note is 75 Hz.

What would be the frequencies of the next two higher notes for this air column?

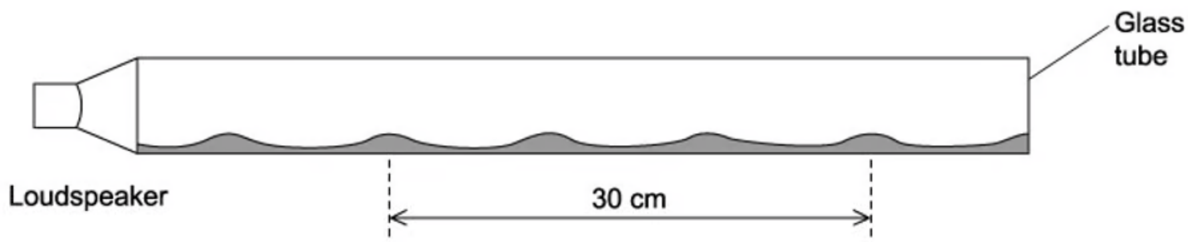
	First higher note / Hz	Second higher note / Hz
A	225	375
B	150	300
C	150	225
D	113	150

[1 mark]

Question 3

A glass tube closed at one end, with a layer of fine powder laid inside as shown in the diagram, was set up. Sound was emitted from the loudspeaker and placed near the open end of the tube.

The frequency of the sound wave was changed, at one frequency a stationary wave was formed, and the powder formed 4 heaps. The distance between the four heaps of powder was 30 cm.



The speed of sound in the tube is 330 m s^{-1}

What is the frequency of the sound?

A. 6600 Hz

B. 1650 Hz

C. 3300 Hz

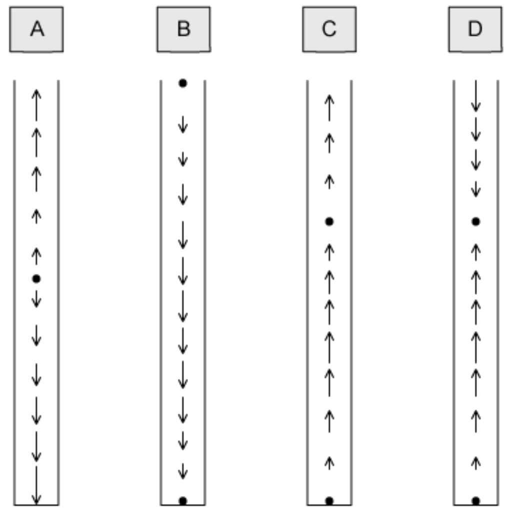
D. 2200 Hz

[1 mark]

Question 4

A pipe is used to set up a stationary longitudinal wave. The length of each arrow in the diagrams below shows the motion of the air molecules. The arrowheads show the direction of the motion of the particle.

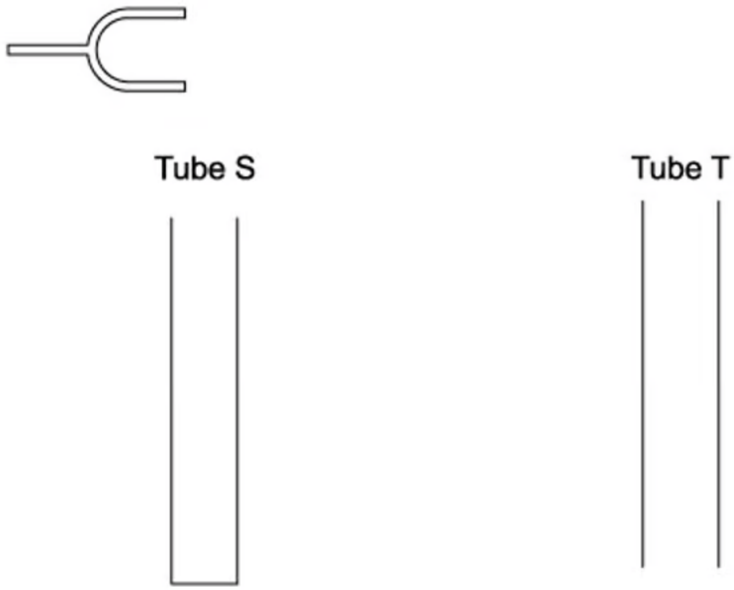
Which diagram shows a stationary wave with two nodes and two antinodes?



[1 mark]

Question 5

A student was investigating a tuning fork over two tubes. The tube were identical in length and size except tube S is closed at one end, and tube T is open at the lower end.



The tuning fork was placed above tube S and caused a resonance of the air at frequency *f*. No resonance is found at any lower frequency than *f* with tube S.

Which tuning fork will produce resonance when placed above tube T?

A. a fork of frequency $2f$

B. a fork of frequency $\frac{3f}{2}$

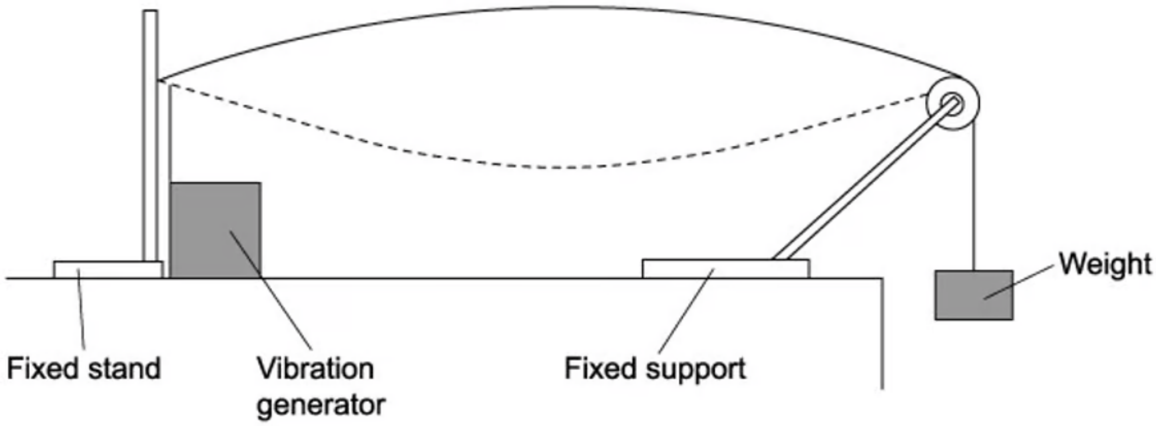
C. a fork of frequency $\frac{2f}{3}$

D. a fork of frequency $\frac{f}{2}$

[1 mark]

Question 6

A steel wire was set up clamped at one end and placed under tension by a mass hung over a pully; this is shown in the diagram.



A vibration generator is attached to the wire nearest to the clamped end. A stationary wave with one loop is produced. The frequency of the generator is *f*.

Which frequency should be used to produce two loops?

A. $2f$

B. $4f$

C. $\frac{f}{2}$

D. $\frac{f}{4}$

[1 mark]

Question 7

An investigation to sound waves reflected by a wall was carried out. The frequency *f* of the wave was adjusted until a stationary wave is formed with an antinode with a distance *x* closest to the wall.

Which expression gives *f* in terms of *x* and the speed of sound c ?

A. $f = \dfrac{c}{2x}$

B. $f = \dfrac{c}{4x}$

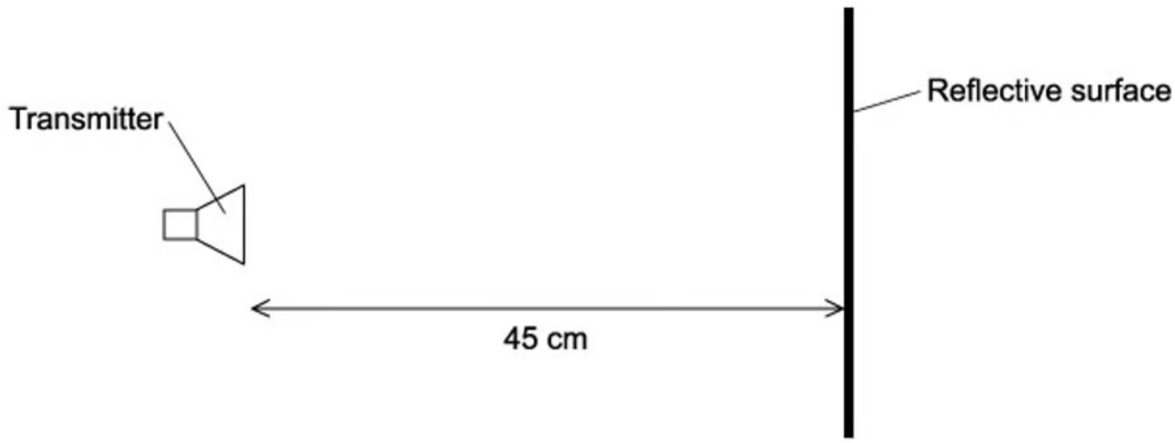
C. $f = \dfrac{2c}{x}$

D. $f = \dfrac{4c}{x}$

[1 mark]

Question 8

A student was investigating the reflection of electromagnetic waves. They set up a transmitter 45 cm from a reflective surface, as shown in the diagram.



The waves had a frequency of 1.00 GHz. A stationary wave was produced with a node at the transmitter and a node at the surface.

How many antinodes are there in the space between the surface and the transmitter?

A. 4

B. 3

C. 2

D. 1

[1 mark]

Question 9

Stationary sound waves can be produced in an organ pipe of length *l* when both ends are open, and this creates a note from an organ.

The speed of sound in the air column is v.

What is the lowest (fundamental) frequency of the note produced in the pipe?

A. $\frac{v}{4l}$

B. $\frac{v}{2l}$

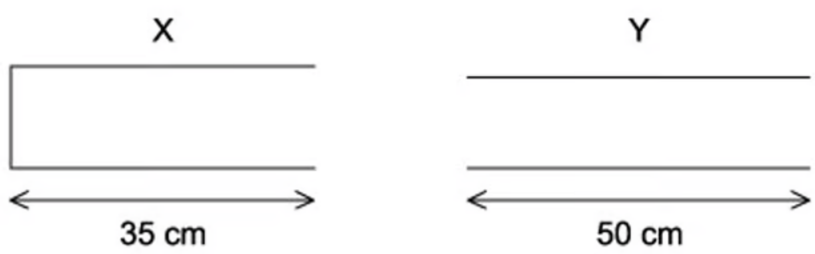
C. $\frac{v}{l}$

D. $\frac{2v}{4l}$

[1 mark]

Question 10

A student was investigating the creation of stationary waves. They set up travelling waves with a wavelength of 20 cm in a closed pipe X and an open pipe Y. The lengths of the pipes are shown in the diagram.



In which pipe did the student observe stationary waves?

A. X only

B. X and Y

C. Y only

D. neither X or Y

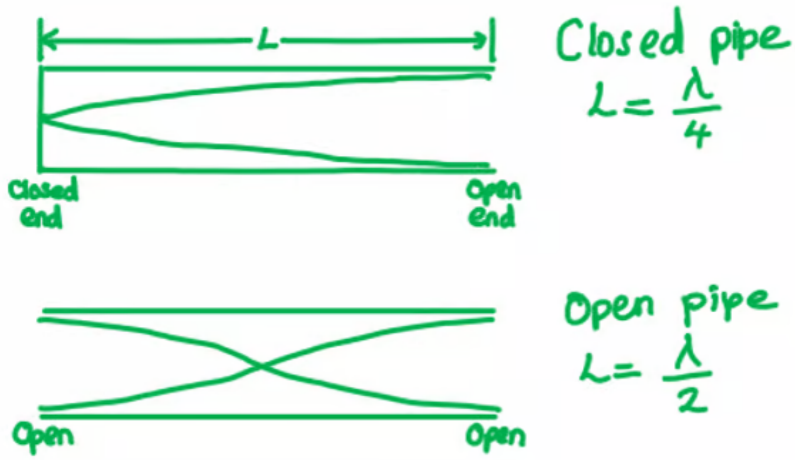
[1 mark]

Model Answers: Hard

1

The correct answer is **A** because:

- For a stationary wave to be produced, there should be a node at the fixed end and an antinode at the loudspeaker
- For the demonstration in the question, the frequency is f and formed in the length of pipe l, for a fundamental mode the wave formed is a quarter of a wavelength
  - For a closed pipe :
  - $L = \frac{\lambda}{4}$ gives a wavelength of $\lambda = 4L$



- Option **A** shows a pipe with length $2L$ with two open ends
  - This would create half a wavelength, for this case, $\frac{\lambda}{2} = 2L$
  - Hence the wavelength is $\lambda = 4L$, which is the same as the demonstration in the question

B is incorrect because a node will form at the closed end of the pipe and an antinode at the speaker, this is a quarter of a wave $\frac{\lambda}{4} = 2L$, therefore, $\lambda = 8L$

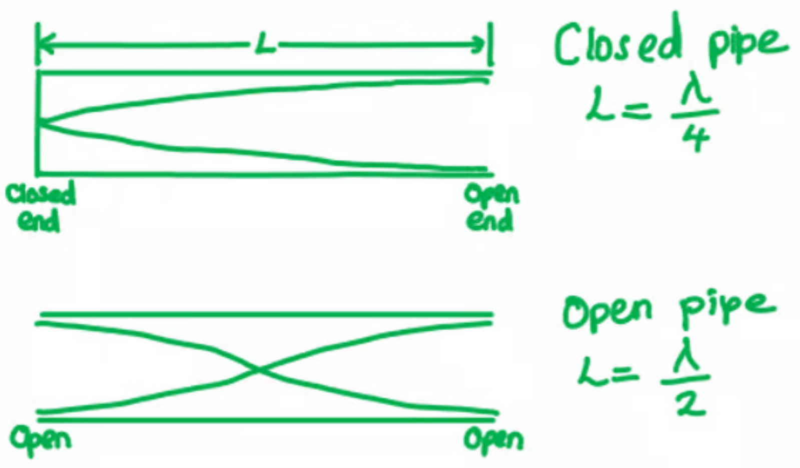
C is incorrect because there will be an antinode at both the speakers giving half a wavelength $\frac{\lambda}{2} = L$, therefore, $\lambda = 2L$

D is incorrect because there would be an antinode at the speaker and an antinode at the open end giving half a wavelength $\frac{\lambda}{2} = L$, therefore, $\lambda = 2L$

2

The correct answer is **A** because:

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- The question states that a node is formed at the mouthpiece and an antinode at the bell of the horn
- The lowest or fundamental note produced will have only one node at the mouthpiece and an antinode at the bell, creating a quarter of a wavelength



- The first higher note will have a node at the mouthpiece, then a node, followed by an antinode at the bell, giving $\frac{3}{4}$ of a wavelength

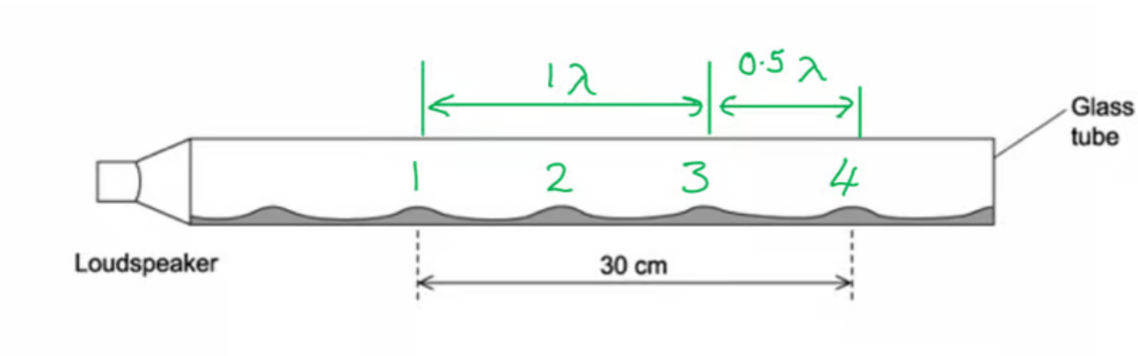
- So, the frequency is 3 times that of the lowest note
- Therefore, frequency $= 3 \times 75 = 225\ Hz$

- The second-highest note would be $\frac{5}{4}$ of a wavelength
  - So, the frequency is 5 times that of the lowest note
  - Therefore, frequency $= 5 \times 75 = 375\ Hz$

3

The correct answer is **B** because:

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- The pile of dust will correspond to a node
- One wavelength is equal to three nodes, so, the distance formed by three successive heaps is equal to one wavelength



- So, the wavelength of the sound wave is :
  - $\frac{3}{2}\lambda = 0.3$
  - $\lambda = 0.3 \times \frac{2}{3} = 0.2\ m$
- The wave equation is given by $v = f\ x$
  - Therefore, frequency, $f = \frac{v}{\lambda} = \frac{330}{0.2} = 1650\ Hz$

4

The correct answer is **D** because:

- A standing or stationary wave is formed from the superposition of two travelling

waves with the same frequency, polarisation and amplitude
- The waves are moving in opposite directions, and the standing wave results from alternately constructive and destructive interference
- A node is represented by a dot on the diagram, which shows zero displacement
- The arrows show the direction of travel of the wave
    - Hence diagram **D** shows two nodes and two antinodes

A is incorrect because the diagram only shows one node

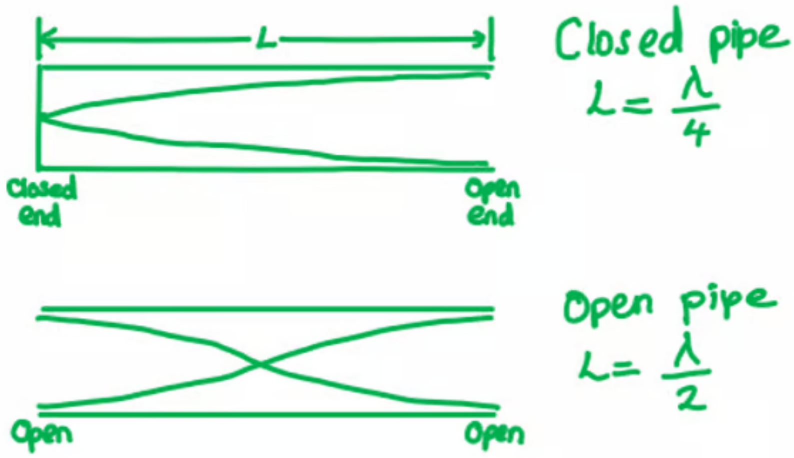
B is incorrect because the diagram only shows one antinode

C is incorrect because a stationary wave is produced by the superposition of the wave travelling in the opposite direction. This diagram shows the waves all travelling in the same direction

5

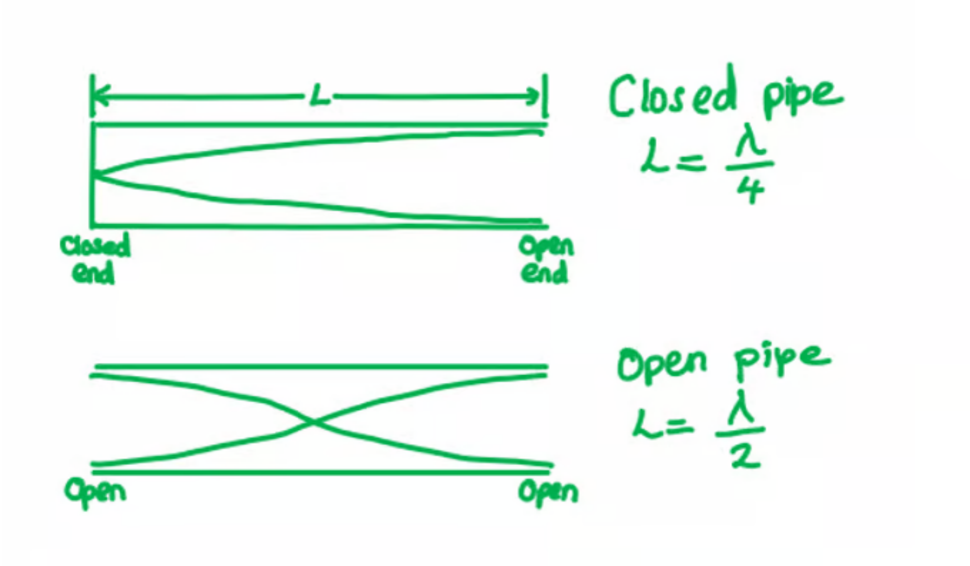
The correct answer is **A** because:

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- Tube S is a close-ended pipe, with a node at the closed-end, and an antinode at the open end
- Tube T is an open-ended pipe, with an antinode at both ends



the open end

- Tube T is an open-ended pipe, with an antinode at both ends



- In a closed tube, the lowest or fundamental note produced will have only one node at the fixed end and an antinode at the open, creating a quarter of a wavelength
  - So $L = \frac{\lambda}{4}$, giving, $\lambda = 4L$
- In an open tube, the lowest or fundamental note produced will have an antinode at each end with a node in the middle, creating half of a wavelength
  - So $L = \frac{\lambda}{2}$, giving, $\lambda = 2L$
- For the closed tube S, the resonant frequencies are equal to:
  - $f_c = \frac{nv}{4L}$
  - Where v = speed of the wave, L = length of the tube and n = 1,3,5...
- For the open tube T, the resonant frequencies are equal to:
  - $f_o = \frac{nv}{2L}$
  - Where n = 1,2,3...
- So, the resonant frequency for the open tube T (when n = 1):
  - $f_o = 2f_c$

6

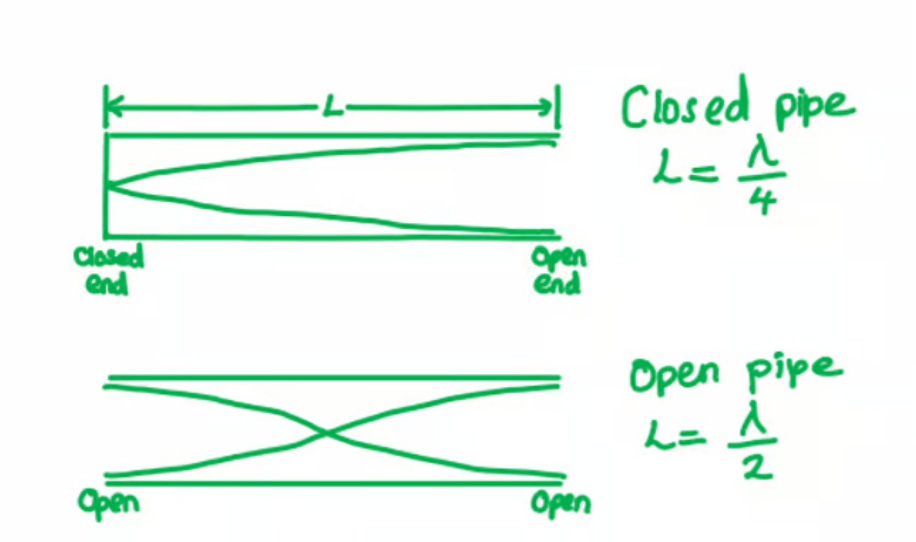
The correct answer is **A** because:

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- In the diagram 1 loop is formed (1st harmonic) and this is half a wave so on the length of the wire, *L*
  - $L = \frac{\lambda_1}{2}$ so, wavelength $\lambda_1 = 2L$
  - Where λ is the wavelength of the wave
- The wave equation is given by $v = f\lambda$
  - So, frequency, $f_1 = \frac{v}{\lambda_1} = \frac{v}{2L}$
- For 2 loops to be formed (2nd harmonic) on the same length of wire:
  - $\lambda_2 = L$, as a complete wave will be formed
- Therefore, the new frequency is equal to :
  - $f_2 = \frac{v}{\lambda_2} = \frac{v}{L} = 2f_1 = 2f$

7

The correct answer is **B** because:

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- A node would be present at the wall, the same as in a closed tube



- The distance between a node and antinode is equal to $\frac{\lambda}{4}$
  - So, $\frac{\lambda}{4} = x$, so, wavelength $\lambda = 4x$
- The wave equation is equal to $c = f\lambda$
  - Therefore, frequency : $f = \frac{c}{\lambda} = \frac{c}{4x}$

8

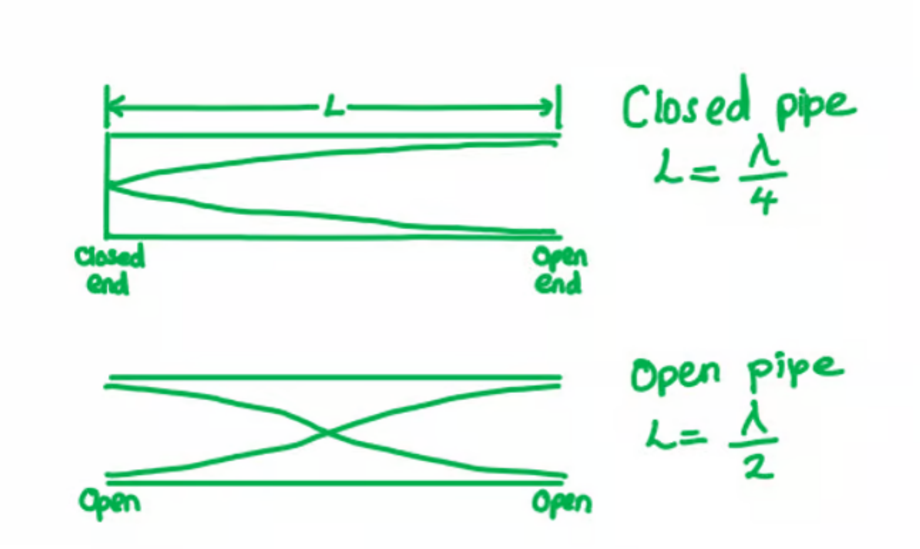
The correct answer is **B** because:

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- The wave equation is equal to $c = \lambda f$
- Where:
  - Speed of electromagnetic waves : $c = 3.0 \times 10^8$ m s^{-1}
  - Frequency : $f = 1$ GHz $= 1 \times 10^9$ Hz
  - Wavelength : $\lambda = \frac{v}{f} = \frac{3.0 \times 10^8}{1 \times 10^9} = 0.3\ m$
- The distance between the transmitter and the reflective surface = 45 cm
  - So, the number of periods = $\frac{0.45}{0.30} = 1.5$
- One antinode is half a period, so, in 1.5 periods there are 3 antinodes

9

The correct answer is **B** because:

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- The organ pipe is open at both ends, in order for it to have the lowest (fundamental) frequency it will have two antinodes, with a single node in the middle



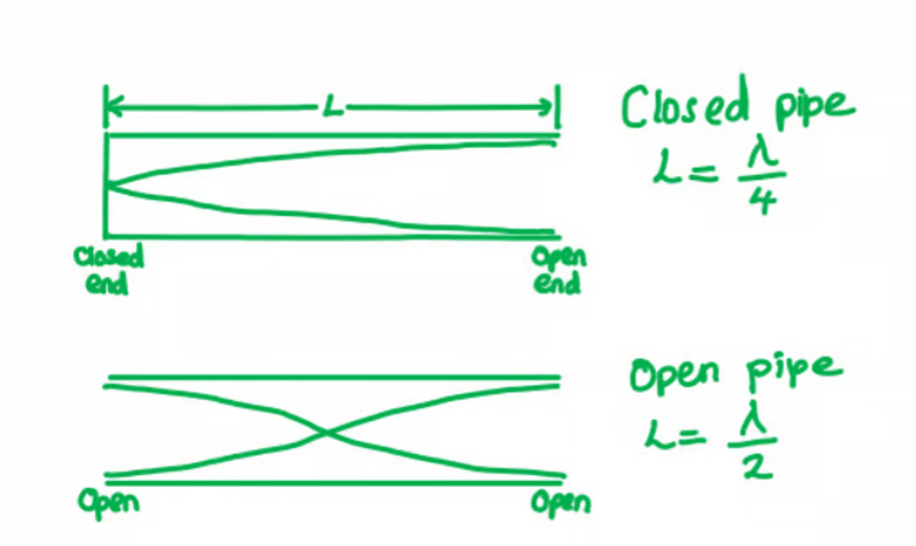
- The wavelength can be measured by measuring the length between 3 consecutive nodes, the distance between the 1st and 2nd nodes give half the wavelength
    - So, in the organ pipe $l = \frac{\lambda}{2}$
    - Speed of a wave, $v = \lambda f$
        - Therefore, the fundamental frequency is $f_f = \frac{v}{\lambda} = \frac{v}{2l}$

10

The correct answer is **B** because:

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude



- In an open-ended pipe there will be half a wavelength, so for pipe Y:
  - $L = \frac{x}{2} \times \lambda$
  - $50 = \frac{x}{2} \times 20$
  - $x = 5$
- There is an odd whole number produced
  - Therefore, a standing wave will be created
- In a close-ended pipe there will be a quarter of a wavelength, so for pipe X:
  - $L = \frac{x}{4} \times \lambda$
  - $35 = \frac{x}{4} \times 20$
  - $x = 7$
- There is an odd whole number produced
  - Therefore, a standing wave will be created
- So, both X and Y will produce a standing wave